



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester MCA (CBCS) Degree Examinations
Summer Semester September 2023
22MCA105 - SOFTWARE ENGINEERING AND TESTING

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing One full question from each Unit

		Unit - I			
		Marks	BT*	CO*	PO*
1.	a) Explain briefly Incremental Software Process Model.	10	L1	1	1
	b) List the stages of Testing followed in Software Validation.	10	L1	1	1
2.	a) Explain briefly Waterfall Model with a neat diagram.	10	L2	1	1
	b) Describe the general Software Process Models.	6	L3	1	1
	c) Mention the four fundamental activities that are common to all Software Processes.	4	L2	1	1
		Unit - II			
3.	a) Differentiate between Functional and Non-Functional Requirements.	10	L1	2	1
	b) Outline the Following				
	i) Interviewing				
	ii) Ethnography	10	L2	2	1
4.	a) Explain the Requirements Elicitation and Analysis process.	10	L3	2	1
	b) List the steps in Requirement Validation.	10	L2	2	1
		Unit - III			
5.	a) With a help of use case diagram explain the role of Medical Receptionist.	7	L2	3	1
	b) With a help of State Diagram explain the working of ATM.	7	L3	3	2
	c) Draw a Sequence Diagram for Patient Information System.	6	L3	3	1
6.	a) What are the several things you need to consider to develop a system design from concept to object-oriented design.	5	L2	3	1
	b) Design an interaction between weather station and weather Information system to report weather data using UML.	8	L3	3	1
	c) Draw a state machine model for weather station.	7	L3	3	2
		Unit - IV			
7.	a) What is White Box Testing. Compare its Methods.	10	L2	4	1
	b) Define the types/goals of system Testing.	10	L1	4	1
8.	a) Explain error guessing and syntax testing.	10	L2	4	1
	b) Describe the lesser used methods in Blackbox Testing.	10	L1	4	1
		Unit - V			
9.	a) Write short note on:				
	i) Unit testing				
	ii) Integration testing	10	L2	5	1
	b) Explain any 2 testing methods in High level Testing.	10	L2	5	1
10.	a) Explain any two Test execution and evaluation tools.	10	L2	5	1
	b) Write short note on				
	i) Testing is each person's responsibility				
	ii) Testing is performed by dedicated resource	10	L3	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
